# Update of country-ocean atmospheric moisture bilateral flows dataset (from v2 to v3)

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**Related repository**: De Petrillo, E., Fahrländer, S., Tuninetti, M., Andersen, L.S., Monaco, L., Thomas F.R., Cigna G., Ridolfi, L., Laio, F. (2026). Country-ocean-moisture-flows-reconciled-with-ERA5-reanalysis obtained processing Lagrangian moisture connections

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This release presents an updated version of the country-scale (including oceans) matrix of bilateral atmospheric moisture-flow connections (Figure 1) based on the reconciliation approach developed in De Petrillo, E., Fahrländer, S., et al., (2025) https://www.nature.com/articles/s43247-025-02289-y applied to the output data of the Lagrangian atmospheric moisture tracking model UTrack by O. A. Tuinenburg and A. Staal (2020). "Tracking the global flows of atmospheric moisture and associated uncertainties". In: *Hydrology and Earth System Sciences* 24.5, pp. 2419–2435. DOI: 10.5194/hess-24-2419-2020. URL: https://hess.copernicus.org/articles/24/2419/2020/.

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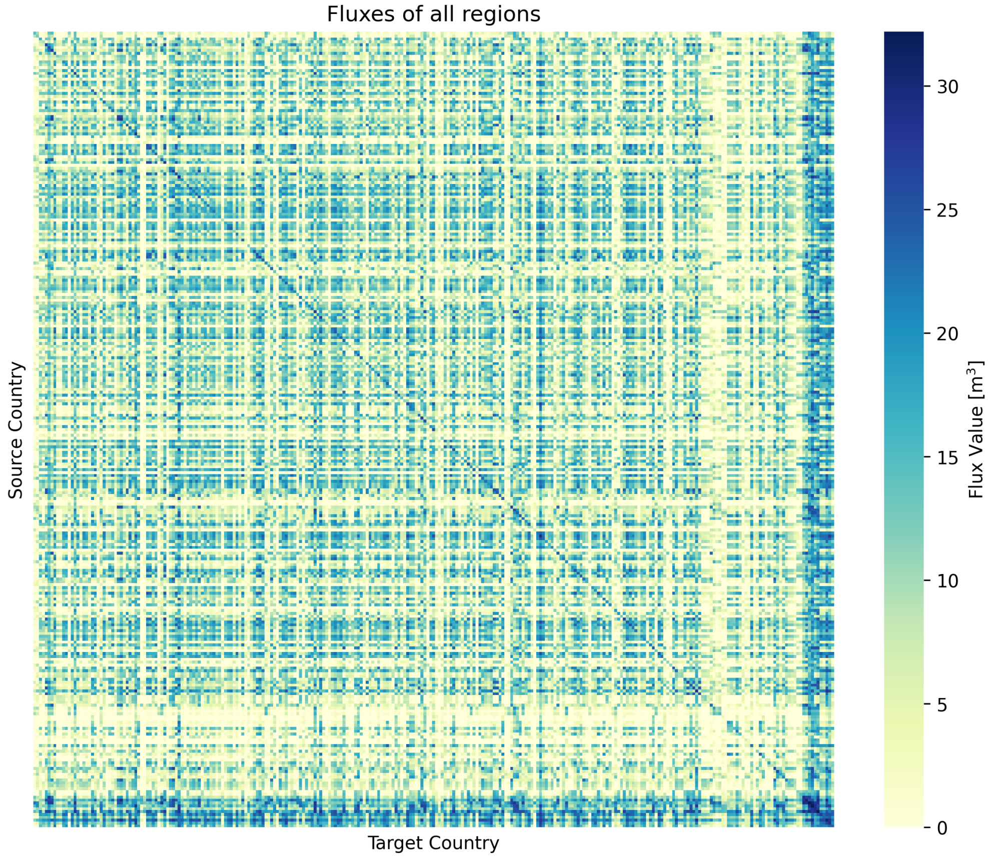


Figure 1: The bilateral matrix describes the flows of water vapor between countries. Its generic element (i, j) - in the i-th row and j-th column - reports the flow evaporated from country i and precipitated over country j. Therefore, across the rows, flows are viewed as evaporation flows, while across the columns, the same flows are viewed as precipitation.

## Update note

The update (v3) pertains to the underlying geographical mask delimiting countries and oceans. This version provides an improved allocation of grid cells (0.5°) over national borders and along coastal boundaries. The moisture flow aggregation from the cell to the country/ocean scale is now based on a 3D mask, which allows us to more precisely attribute water volumes associated with cells that span country borders or extend along coasts. The 3D mask is a set of $n$ 2D masks, one for each country/ocean (3rd dimension; see Figure 2), and each cell contains a fractional value between 0 and 1. This value corresponds to the share of the cell that is in the country/ocean, with 1 indicating that the cell (defined by longitude and latitude) is fully in the country/ocean. Further details on the 3D mask methodology, data, and procedures are available at https://github.com/romain894/region-mask (Romain François Thomas and Giulia Cigna (2026). *3D Fractional Region Mask for Aggregating Global Gridded Data*. DOI: 10.5281/zenodo.19007672. URL: https://doi.org/10.5281/zenodo.19007672)

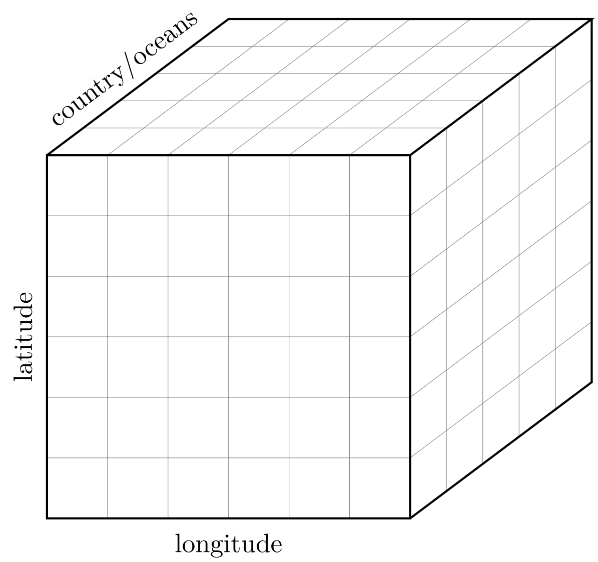


Figure 2: 3D mask structure: The first dimension of the mask is the country/ocean (region). For each region, a 2D array with a size of latitude × longitude stores the fractional value of each cell area in the region.

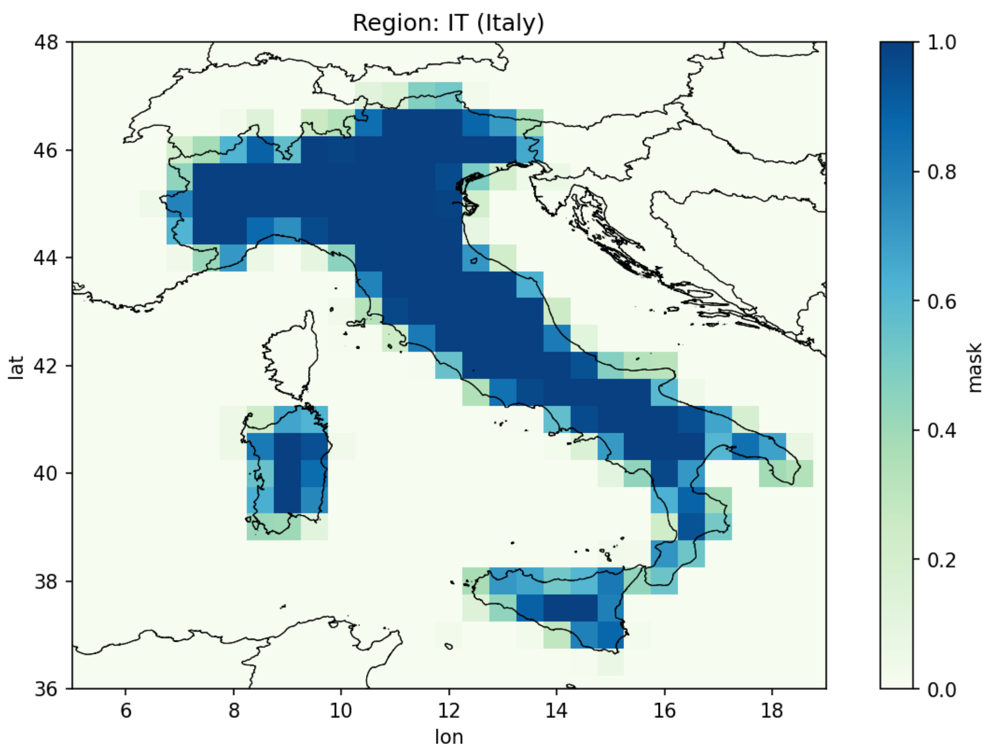


Figure 3: Example of the updated geographical mask. The map shows the fractional values of the 2D array for Italy. The cells fully in Italy have a value of 1 (blue) while the cells completely outside of Italy have a value of 0 (light green). The cells on the coastlines and borders store the fraction of the cell area within Italian borders.

# Data

The country-ocean atmospheric moisture bilateral flows matrix was created using the following geographic delineations:

**Eurostat (edited) / GOaS / SeaVoX:**

**Countries 2020** (Eurostat)
https://ec.europa.eu/eurostat/web/gisco/geodata/administrative-units/countries

**Global Oceans and Seas** version 1 (Flanders Marine Institute, 2021)
https://doi.org/10.14284/542

**Polygon dataset of the extent of water bodies from the SeaVoX Salt and Fresh Water Body Gazetteer** v19 (British Oceanographic Data Centre, 2023)
https://doi.org/10.14284/590

Those three shape files were combined in a single shape file, which was used as the input to create the 3D mask.

The editing of the Eurostat delineation consists of separating overseas or detached territories from the mainland. This is the case of all overseas French territories, Hawaii from the United States, Taiwan from China, and Alaska from the United States.

**Subcontinental atmospheric connection matrix:**

The repository also includes a subcontinental matrix of atmospheric connections between subcontinental regions, including oceanic areas.

The delineations used in this dataset align with those in the Eurostat (edited), GOaS, and SeaVoX analyze. The only difference lies in the aggregation of countries, which follows the following classification:

**Geographical Regions Classification (M49)** (United Nations Statistics Division (UNSD))
https://unstats.un.org/unsd/methodology/m49/